

Mohamed A. Abdelraouf

AI & Embedded Engineer · TinyML · Robotics · PCB Design

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PROFILE

Engineer working where software meets hardware — building shipped AI agents, TinyML models for edge devices, and custom robotics/PCB hardware end to end. Comfortable across the full stack: model training and quantization, embedded firmware, electronics design, and fabrication. Open to roles in AI, TinyML, embedded systems, and robotics.

SELECTED PROJECTS

Local Tab Agent + Swarm Compute

AI AGENT · SHIPPED

Local AI coding agent for VS Code, published on the Open VSX Registry. Repository-aware planning, browser AI-tab bridging (ChatGPT / Gemini / Claude), and distributed "Swarm Groups" for GPU/CPU pooling.

650+ installs · 3 AI bridges · distributed compute — TypeScript, Ollama, Local LLMs, WebSocket, VS Code API

Custom Wake-Word Engine (KWS)

TINYML · SPEECH

Keyword-spotting pipeline that trains custom wake words for edge devices from small datasets with heavy audio augmentation, exported to TensorFlow Lite.

~98% accuracy · TF Lite export · edge-ready — TensorFlow, TF Lite, Librosa, TinyML

Panic Attack Detection System

MEDICAL ML · WEARABLE

Real-time panic-attack detector fusing multi-sensor wearable inputs (heart rate, EDA, temperature, motion) with an ML ensemble and personalized baseline calibration.

~1s detection · 4 sensors fused · personalized baseline — Python, scikit-learn, Random Forest, Arduino

EMG Bionic Hand

ROBOTICS · BIOMEDICAL

Prosthetic robotic hand driven by EMG muscle/brain signals processed in real time into multi-servo motion. Presented at the Pharos University exhibition.

EMG-driven · multi-servo · exhibited — Arduino, EMG sensors, servos, Embedded C

Schematic Sidekick

AI · HARDWARE DESIGN TOOL

AI tool that turns a project goal into a hardware design — suggesting boards/modules, then handling wiring and firmware logic. Goal to schematic in minutes.

goal → schematic · auto-wiring — AI, electronics, schematic generation, firmware

Micropad — Senet Labs

EMBEDDED · PCB · STARTUP

Production-ready macropad controlling editing/3D software, media, apps and sites, with a customizable Electron companion app and OTA firmware updates.

USB HID · OTA · STM32F4 — STM32F4, C, KiCad, USB HID, Electron, React

ALSO BUILT

- Self-built FDM 3D printer — frame, motion, wiring & firmware from scratch
- DIY PCB fabrication lab — electroless copper plating
- AI Desktop Assistant — voice + vision, bilingual AR/EN (PyQt6, Gemini, OpenCV)
- Interview Booster — real-time STT copilot with local Whisper inference

CERTIFICATIONS

Harvard University — CS50: Introduction to Computer Science

HarvardX — Fundamentals of TinyML

HarvardX — Applications of TinyML

Edge Impulse — Intro to Embedded Machine Learning

CORE SKILLS

Python

TypeScript

Embedded C

TensorFlow

TF Lite

scikit-learn

TinyML

Local LLMs / Ollama

OpenCV

STM32

Arduino

KiCad

PCB Design

3D Printing / CAD

Electron

React

DeepLearning.AI — TensorFlow Developer

DeepLearning.AI — Neural Networks & Deep Learning

DeepLearning.AI — Improving Deep Neural Networks

LinkedIn Learning — Deep Learning: Model Optimization